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Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE - Academic Year 2026-27

Java Programming – Lab (Batch 2025-29)

Lab Assignment No:- 14

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Batch	2025-29
Class	Programming with JAVA
Academic Year & Semester	2nd Year – 3rd Sem
Title of Assignment	Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks
Practice questions	1. Create an ATM program handling invalid withdrawal amount using try-catch.

```java
public class ATMdemo {
    public static void main(String[] args) {
        double balance = 5000.0;
        double withdrawAmount = 7000.0;

        try {
            if (withdrawAmount > balance) {
                throw new IllegalArgumentException("Insufficient balance for
withdrawal");
            }
            balance -= withdrawAmount;
            System.out.println("Withdrawal successful. Remaining balance: " +
balance);
        } catch (IllegalArgumentException e) {
            System.out.println("Transaction failed: " + e.getMessage());
        }
    }
}
```

	2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.
	
	`public class Onlineshoppingdemo {`
	`  public static void main(String[] args) {`
	`  int quantity = 0;`
	
	`  try {`
	`  if (quantity <= 0) {`
	`  throw new IllegalArgumentException("Quantity must be greater than zero");`
	`  }`
	`  System.out.println("Order placed for " + quantity + " item(s)");`
	`  } catch (IllegalArgumentException e) {`
	`  System.out.println("Order failed: " + e.getMessage());`
	`  }`
	`  }`
	`}`

Output	1)
	
	`eabe1187\bin' 'JAVA_EXP_14.ATMdemo'`
	`Transaction failed: Insufficient balance for withdrawal`
	`PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>`
	
	2)
	
	`va\jdt_ws\JAVA_PRACTICALS_eabe1187\bin' 'JAVA_EXP_14.Onlineshopping`
	`Order failed: Quantity must be greater than zero`
	`PS C:\Users\abell\OneDrive\Desktop\JAVA_PRACTICALS>`

Conclusion	Exception handling allows a Java program to detect and respond to runtime errors gracefully instead of crashing. Multiple catch blocks let a single try block respond differently depending on the exception type, while nested try-catch supports handling errors at different levels of a program's logic — together making applications more robust and user-friendly.